

Infinitely Many Closed Geodesic Images on Reversible Finsler Spheres

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Theorem 1. *Let $n \geq 3$. Every smooth, strongly convex, reversible Finsler metric on the standard sphere S^n has infinitely many geometrically distinct prime closed geodesics.*

Here smoothness is required off the zero section, and strong convexity means that the fiber Hessian of $F^2/2$ is positive definite at every nonzero tangent vector. Geometric distinctness means distinct images in S^n ; iteration, a change of starting point, and reversal do not produce additional images.

1 Notation and normalization

Fix such a metric F . Put $d = n - 1$, $\Lambda = H^1(S^1, S^n)$, and let $M \subset \Lambda$ be the constant loops. All homology and cohomology groups have coefficients in $\mathbb{F}_2$. Set

$$E(\gamma) = \int_0^1 F(\gamma, \dot{\gamma})^2 dt, \quad \ell(\gamma) = \int_0^1 F(\gamma, \dot{\gamma}) dt, \quad X_a = \{\sqrt{E} < a\}, \quad \mathcal{A}_a = \{\ell < a\}.$$

Let $r\gamma(t) = \gamma(-t)$, $\tau\gamma(t) = \gamma(t+1/2)$ and $\delta\gamma(t) = \gamma(2t)$. Write $R = \langle r \rangle$, $K = \langle r, \tau \rangle$ and $G = O(2)$, acting by changes of parameter. Reversibility and homogeneity give

$$E(r\gamma) = E(\gamma), \quad E(\delta\gamma) = 4E(\gamma), \quad X_{2a}^\tau = \delta X_a.$$

For an invariant pair, the subscript on H_H^* denotes Borel cohomology. We use

$$A_a^j = H^j(X_a, M), \quad B_a^j = H_R^j(X_a, M),$$

and use $a = \infty$ for the full loop space. All negative-degree groups are zero. Constants denoted by C may increase between estimates; none depends on a varying cut or degree.

For a nonconstant loop put $v(t) = F(\gamma(t), \dot{\gamma}(t))$ and $L = \int_0^1 v$. For $0 \leq u < 1$ define

$$\phi_u(t) = (1-u)t + \frac{u}{L} \int_0^t v(s) ds, \quad \Psi_u \gamma = \gamma \circ \phi_u^{-1}.$$

At $u = 1$, collapse each interval on which the integral is constant. The loop is constant on such an interval, so the resulting loop is well defined. Constant loops are fixed. Change of variables gives

$$E(\Psi_u \gamma) = \int_0^1 \frac{v(t)^2}{1-u+uv(t)/L} dt \leq (1-u)E(\gamma) + uL^2 \leq E(\gamma), \quad E(\Psi_1 \gamma) = L^2. \quad (1)$$

For the inequality, multiply

$$(1-u)s^2 + us - \frac{s^2}{1-u+us} = \frac{u(1-u)s(s-1)^2}{1-u+us} \geq 0$$

by L^2 and integrate with $s = v/L$.

This is a strong H^1 -continuous homotopy. Indeed, for $\gamma_i \rightarrow \gamma$ in H^1 and $u_i \rightarrow u$, the speeds converge in L^2 and the cumulative integrals converge uniformly. Taking limits of inverse fibers shows uniform convergence of the output loops, including $u = 1$. Their velocities are bounded in L^2 , and every weak limit is the velocity of the uniform limit. The integral formula gives energy convergence if $u < 1$; if $u = 1$, use

$$L_i^2 \leq E(\Psi_{u_i} \gamma_i) \leq (1 - u_i)E(\gamma_i) + u_i L_i^2 \longrightarrow L^2.$$

In coordinates, compactness and strong convexity give, uniformly in the base point,

$$F_x(z)^2 \geq F_x(y)^2 + d_v(F_x^2)(y)[z - y] + c|z - y|^2.$$

Move the velocities to the limiting tangent spaces by orthogonal projection in an auxiliary embedding. This changes their L^2 norms and energies by quantities tending to zero. Integrating the displayed inequality, weak convergence kills the linear term and energy convergence forces strong L^2 convergence. At a constant limiting loop, (1) gives the same conclusion directly.

2 A uniform filling bound

Lemma 1. *There is $D > 0$ such that, for every $a > 0$ and every $j \geq 0$,*

$$\ker(H_j(X_a, M) \rightarrow H_j(\Lambda, M)) = \ker(H_j(X_a, M) \rightarrow H_j(X_{a+D}, M)).$$

Proof. First fix j . A bounded-length path block can be replaced locally, relative to its endpoints, by a path of uniformly bounded length with uniformly bounded additional length. To see this, divide it into pieces of length at most one. Choose short spokes from its initial point to the division points. Replace a spoke followed by the next piece by the next spoke, successively. The path pairs used at a single step have a common Lipschitz bound. Their space is compact in C^0 . Simple connectivity supplies a relative-endpoint homotopy at each pair; polygonal approximation and short endpoint connectors extend it to a neighborhood, continuously in H^1 . A finite cover supplies a common length bound. Only the active prefix changes, so its overhead does not accumulate with the number of pieces.

For a finite parameter polyhedron of dimension j , normalize its loops and split them into $k = \lfloor a/B \rfloor$ equal time blocks. Choose B so that $B/2 > D_0 + jC_0 + 2$, where D_0 is the terminal block bound and C_0 its overhead. Subdivide the parameter polyhedron and color its barycentric vertices by dimensions $0, \dots, j$. A vertex of color h applies its local recipe to block $h + 1$, for the fraction $\min\{1, (j + 1)\lambda_v\}$ at terminal time. Recipes of the same color have disjoint positive-coordinate supports; different colors change different blocks. Thus they paste continuously. At least one block is completely replaced. For a loop of length L , the terminal length is at most

$$L - L/k + D_0 + jC_0.$$

If $L \geq a/2$, this is less than $a - 1$. If $L < a/2$, the bound $L + (j + 1)C_0$ has the same consequence for sufficiently large a . The homotopy stays below $a + (j + 1)C_0$.

A fixed bounded level has a single compact finite polygon model: normalize, sample at sufficiently many equal times, and interpolate using an auxiliary smooth Riemannian metric. A regular discrete-energy sublevel containing the sampled loops is a compact manifold with boundary. The interpolation homotopy has uniformly bounded F -length. The kernel of its degree- j homology map to Λ is finite-dimensional. Choosing fillings of a basis gives one bounded level in which every globally null class from this model dies. Repeated one-unit shortening, followed by these fillings, therefore gives a constant D_j such that a globally null j -cycle in $\mathcal{A}_a$ bounds in $\mathcal{A}_{a+D_j}$. The successive height bounds are $a - r + (j + 1)C_0$, so they are not added. A finite singular

cycle is handled by gluing its cancelling faces and subdividing to a finite parameter polyhedron; the subdivision prisms preserve images.

We make this bound independent of j . Write $\bullet$ for the degree- $-n$ loop homology product and $\otimes$ for the degree- $(n-1)$ relative cohomology product. The sphere product and resonance identities give $\alpha > 0$, C_H , and classes

$$\Theta \in H_{3n-2}(\Lambda), \quad \omega \in H^{n-1}(\Lambda, M), \quad \theta \in H^{3n-2}(\Lambda, M)$$

such that the ordinary absolute groups have rank at most one in each degree and

$$|\Theta^{\bullet m}| = n + 2md, \quad |\omega^{\otimes m}| = (2m-1)d, \quad \langle \theta \otimes \omega^{\otimes m}, \Theta^{\bullet(m+1)} \rangle = 1, \quad (2)$$

$$|\deg z - \alpha \operatorname{cr}(z)| \leq C_H \quad \text{for each nonzero homogeneous ordinary class } z. \quad (3)$$

Here cr is the least carrying level in homology, and the supremum of vanishing restriction levels in cohomology. We use the relative cohomology statement above the degrees of the constant sphere.

Put $q = 2md$, $p = q - d$, and choose representatives

$$S = \Theta^{\bullet m} \text{ in } \mathcal{A}_l, \quad T = \omega^{\otimes m} \text{ vanishing on } \mathcal{A}_L,$$

where

$$l = \frac{n+q+C_H}{\alpha} + 2, \quad L = \frac{p-C_H}{\alpha} - 2 > 0, \quad G_0 = l - L = \frac{2n-1+2C_H}{\alpha} + 4.$$

Take m large enough that $p > n+1$. A cocycle relative to X_L , pulled back by Ψ_1 , supplies the required normalized T .

Here is the chain comparison we need. Use a fixed round metric for short arcs and tubes. The diagonal Thom cocycle U_c may be chosen relative both to the complement of a small diagonal tube and to the diagonal itself: its restriction to the diagonal is the mod-two Euler class of TS^n , which is zero. For nearby endpoints set

$$C_1(x, t) = x[0, t] \# [x(t), x(0)], \quad C_2(x, t) = [x(0), x(t)] \# x[t, 1].$$

Concatenations use positive half-times; the maps are H^1 -continuous also at collapsing pieces. On sufficiently small chains define

$$D_T x = (C_2)_\#((x \times [0, 1]) \cap U_c(x(0), x(t)) \cap T(C_1(x, t))).$$

Its degree is $-q$. The time-end terms vanish because U_c is relative to the diagonal. The cap boundary identity proves that D_T takes cycles to cycles and globally null cycles to globally null cycles. If T is nonzero on a retained face, subdivision makes the first-cut length at least $L - \varepsilon$ on its whole parent. The two added arcs have total length less than 2ε ; hence

$$D_T : C_*(\mathcal{A}_a) \longrightarrow C_{*-q}(\mathcal{A}_{a-L+1}). \quad (4)$$

The join μ is cap product with a diagonal Thom cocycle at the starting points followed by concatenation. Before concatenating, rotate the first loop by the shortest round rotation taking its start to the second start. It induces $\bullet$ and satisfies the boundary formula. After any finite collection of input length bounds is fixed, its radius can be chosen so that its length cost is less than one. Indeed

$$\sup_{F(x,v)=1} |F(Jx, dJv) - 1| \longrightarrow 0 \quad (J \rightarrow 1).$$

For a constant fundamental cycle c , normalized shuffle chains give $\mu(c, x) \simeq x$ without increasing length: all surviving constant-factor steps occur in the Thom face, with coefficient one, and the output is x preceded by a constant wait.

For cycles x, y of degrees greater than $n + p$, the same operations give

$$\mu(D_T x, y) \simeq \mu(D_T y, x) \quad \text{in } \mathcal{A}_{a+b-L+4} \quad (x \subset \mathcal{A}_a, y \subset \mathcal{A}_b). \quad (5)$$

To verify this, cut the joined loop $(Jx) \# y$ using the full cut-time interval. In its second half the cut evaluation and retained output depend only on y and time. The join face uses at most n of the x -steps, and the test face at most p . An unused x -step therefore makes either the cut Thom face or the output degenerate. This half vanishes modulo degenerate chains. In the first half, change the durations to those of $(JC_2(x, t)) \# y$ and continuously remove the rotation from the tested first cut, keeping this output fixed. Commute the join cap and the cut/test cap by

$$\delta(\sigma \smile_1 \gamma) = \sigma \smile \gamma + \gamma \smile \sigma + \sigma \smile_1 \delta\gamma, \quad \gamma = U_c \smile T(C_1).$$

This identifies the first half with $\mu(D_T x, y)$. Rotating the entire joined loop back and shifting its parameter by a half-period interchanges x, y . The same argument then gives $\mu(D_T y, x)$.

For these cap computations, extend the tube-defined pullback of T by zero, and subdivide so that a nonzero Thom face puts its entire parent product rectangle inside the larger tube. The identity $\delta(U_c \smile \widetilde{h^*T}) = U_c \smile \delta\widetilde{h^*T}$ is zero there. Apply the same parent condition to the block and skip faces in $\smile_1$ and to the comparison prisms. A nonzero T -face supplies a point of length at least L ; the oscillation of the continuous length function on the parent is less than ε . Thus every comparison retains the same length saving. Choose the join radius after all finite length bounds; the cut radius is already fixed. The short arcs and rotations cost less than one, and the allowance four covers all comparisons. A sum of chains takes a union of supports and does not add their length bounds.

For a relative cocycle b , the cap definition also gives

$$\langle b, D_T z \rangle = \langle [T] \otimes [b], [z] \rangle.$$

Both tests are relative to constants, so changing the Thom representative produces no time-end contribution. Put $Z = \Theta^{\bullet(m+1)}$. Equations (2) and the rank-one calculation give $[D_T Z] = \Theta$. Applying (5) to Z, S gives

$$0 \neq \Theta^{\bullet(m+1)} = [\mu(D_T Z, S)] = [\mu(D_T S, Z)].$$

Since $H_n(\Lambda) = \mathbb{F}_2[M]$, it follows that $[D_T S] = [M]$.

For a sufficiently large degree j , choose

$$m = \left\lfloor \frac{j-2}{2d} \right\rfloor, \quad r = j - q \in \{2, \dots, 2n-1\}.$$

If x is a globally null j -cycle in $\mathcal{A}_a$, choose

$$\partial W = D_T x, \quad W \subset \mathcal{A}_{a-L+1+D_r}, \quad \partial V = D_T S + c, \quad V \subset \mathcal{A}_{G_0+1+D_n}.$$

If the cut level is nonpositive, take $W = 0$. Choose chains H, U_0 with

$$\partial H = \mu(D_T x, S) + \mu(D_T S, x), \quad \partial U_0 = \mu(c, x) + x.$$

Then

$$B_x = \mu(W, S) + \mu(V, x) + H + U_0, \quad \partial B_x = x.$$

Take D_* to be the maximum of the finitely many low-degree filling constants, including all remainder degrees and all degrees below the threshold for this construction. Every term is supported below $a + G_0 + D_* + 6$. Thus one $D = G_0 + D_* + 6$ works in all degrees.

Apply Ψ_1 to a length filling and add the normalization prism of its boundary. Equation (1) gives the same bound for X_a . Finally the chain map $P = 1 - i_{\#}(\text{ev}_0)_{\#}$ kills constants and preserves sublevels. It takes a relative cycle to an absolute cycle and a globally relative-null cycle to an absolutely null one. Applying the absolute bound to P proves the lemma. $\square$

3 Equivariant classes

Let w, U be the first and second Stiefel–Whitney classes of the standard G -plane; let v be the character of K which is one on τ and zero on r . Then

$$H^*(BG) = \mathbb{F}_2[w, U], \quad H^*(BK) = \mathbb{F}_2[v, w], \quad U|_K = v(v + w), \quad |w| = |v| = 1, \quad |U| = 2.$$

The relative sphere-bundle sequences give, naturally for pairs,

$$\cdots \rightarrow H_G^{j-2} \xrightarrow{U} H_G^j \rightarrow H_R^j \rightarrow H_G^{j-1} \xrightarrow{U} H_G^{j+1} \rightarrow \cdots, \quad (6)$$

$$0 \rightarrow H_G^j \rightarrow H_K^j \rightarrow H_G^{j-1} \rightarrow 0, \quad (7)$$

$$\cdots \rightarrow B^{j-1} \xrightarrow{w} B^j \rightarrow A^j \xrightarrow{\text{Tr}} B^j \xrightarrow{w} B^{j+1} \rightarrow \cdots. \quad (8)$$

There is likewise a sign sequence for $K \rightarrow R$ with multiplication by v . The middle sequence uses the plane representation doubling rotation angles; its mod-two Euler class is zero. The first and last use the standard plane and the sign line. These constructions commute with restriction and cup product. Put $e = \text{ev}_0^*[S^n] \in H_R^n(\Lambda)$.

Lemma 2. *For $a_m = (2m - 1)d$, there are classes $X_m \in B_\infty^{a_m}$ such that*

$$\{w^k X_m, ew^k X_m : m \geq 1, 0 \leq k \leq d\}$$

is a graded basis, and $w^n B_\infty = 0$. Each X_m has a lift $\zeta_m \in H_G^{a_m}(\Lambda, M)$. If m is odd, $U^n \zeta_m = 0$. If m is even, eX_m also has a G -lift.

Proof. Filter the same loop space using a round metric. Its m -th positive critical manifold is the space of orthonormal two-frames, with negative-bundle rank a_m . The equivariant crossing is its Thom module shifted by a_m . Reflection sends (x, u) to $(x, -u)$, so its quotient is $P(TS^n)$. Since $TS^n \oplus \mathbb{R}$ is trivial, the projective-bundle formula gives

$$H_R^*(C_m) = \mathbb{F}_2[w, e]/(w^n, e^2).$$

Write x_m for the crossing bottom. For odd m , the full-group critical Borel space maps to $\text{Gr}_2(\mathbb{R}^{n+1})$ with fiber $B(\mathbb{Z}/m)$, which is mod-two acyclic in positive degrees. Define

$$A_0 = 1, \quad A_1 = w, \quad A_i = wA_{i-1} + UA_{i-2}.$$

The complementary bundle on this Grassmannian has rank d and total Stiefel–Whitney class $(1 + w + U)^{-1}$. Consequently the odd crossing is generated by x_m with $A_{d+1}x_m = A_{d+2}x_m = 0$. For even m , τ acts trivially on C_m , and the K -crossing is the reflection Thom module tensored with $\mathbb{F}_2[v]$. The monic polynomial $v(v + w)$ is injective on it. Sequence (7) makes U injective on the G -crossing too. Sequence (6) then shows that this module is generated by x_m and a lift y_m of ex_m : generation modulo U implies generation by induction down the degree.

Let Z_p contain the first p round critical manifolds. Every crossing sequence for (Z_p, Z_{m_0-1}) splits into short exact sequences of vector spaces. For R , its connecting map vanishes on each earlier bottom by degree, and hence on its w, e -multiples. For G the same holds for bottoms and even tops. A top could meet the next bottom only if $n = 3$ and the previous stage is even. Restriction to the one-dimensional reflection bottom is injective, and naturality with the already vanishing reflection connecting map settles this case. This proves the assertion by induction on p .

It follows that a crossing class of degree $j < a_{m+1}$ extends uniquely to $H_H^j(\Lambda, Z_{m-1})$, for $H = R, G$. Indeed, subsequent crossings have increasing bottom degrees; finite singular chains have bounded round energy, so the stabilized degreewise groups compute the full pair. Extend the bottoms in this way. Extend the even tops as well, since $n < 2d$. The same inequality extends

the local relation $w^n x_m = 0$ to $w^n X_m = 0$. The short exact sequences now give precisely the basis in the statement; independence can also be checked at the first round stage occurring in a finite relation.

For odd m , the recurrence gives

$$A_i^2 + A_{i-1}A_{i+1} = U^i.$$

Locally $UA_d x_m = 0$ and $A_d x_m \neq 0$, the latter because its reflection restriction is $w^d x_m$. The source of integration in (6) in degree $a_m + d + 1$ is spanned by ex_m . Exactness therefore gives $\int ex_m = A_d x_m$. For the supported extension $\tilde{\zeta}_m$ this identity extends globally in offset $d < 2d$, and hence

$$UA_d \tilde{\zeta}_m = 0, \quad A_{d+1} \tilde{\zeta}_m = 0.$$

Multiply the polynomial identity at $i = d$ by $U\tilde{\zeta}_m$ to obtain $U^{d+1}\tilde{\zeta}_m = 0$. This gives $U^n \zeta_m = 0$. In dimension three the two relations are $U(w^2 + U)\tilde{\zeta}_m = 0$ and $w^3 \tilde{\zeta}_m = 0$; their extensions use offsets two and three, both below the next bottom at offset four. $\square$

4 Detection in a bounded degree band

Lemma 3. *There is $\mathcal{C}$ such that, for $j \geq 2n + 1$,*

$$B_\infty^j \longrightarrow B_a^j \quad \text{is} \quad \begin{cases} \text{injective,} & j < \alpha a - \mathcal{C}, \\ \text{zero,} & j > \alpha a + \mathcal{C}. \end{cases}$$

Proof. By field duality and Lemma 1, if $b \geq a + D$, then

$$\text{im}(A_b^q \rightarrow A_a^q) = \text{im}(A_\infty^q \rightarrow A_a^q). \quad (9)$$

Both are the annihilator of the same homology kernel; linear functionals extend from subspaces, so finite-dimensionality is unnecessary here.

Put $K_a^q = \ker(B_\infty^q \rightarrow B_a^q)$. If the ordinary restriction in degree q is injective at $b \geq a + D$, then

$$K_b^q \subset wK_a^{q-1}. \quad (10)$$

For $x \in K_b^q$, (8) gives $x = wy$ globally. At b one has $y_b = \text{Tr } z_b$. Choose z_∞ with the same restriction at a as z_b , using (9). Replacing y by $y + \text{Tr } z_\infty$ leaves $wy = x$ and makes $y_a = 0$.

More generally, if $w^N B_\infty = 0$ and ordinary restriction at a is injective in degrees $j - N + 1, \dots, j$, iteration of (10) gives

$$K_{a+(N-1)D}^j \subset w^{N-1} K_a^{j-N+1} \subset w^N B_\infty^{j-N} = 0.$$

For the zero statement, suppose ordinary restriction is zero at $b = a + (N - 1)D$ in degrees $j, \dots, j + N - 1$. Given $x \in B_\infty^j$, nilpotence and (8) express $w^{N-1}x$ as a global transfer, so it is zero at b . If $w^t x_{a+tD} = 0$, exactness expresses $w^{t-1}x_{a+tD}$ as a transfer. Restrict to $a + (t - 1)D$ and use (9); its source becomes a global ordinary restriction, which is zero. Descending to $t = 1$ gives $x_a = 0$.

Use $N = n$, Lemma 2, and (3). One may take

$$\mathcal{C} = C_H + \alpha(n - 1)D.$$

The least degree required on the injective side is $j - n + 1 \geq n + 2$, so the ordinary comparison stays above the constant-sphere degrees. $\square$

5 Finite critical sets and local comparisons

Assume, for a contradiction, that only finitely many prime images occur. Choose representatives $c_1, \dots, c_p$, with lengths $\ell_i > 0$. Every nonconstant closed geodesic has a least positive period. Two prime geodesics with the same image agree, after unit-speed parametrization, up to translation and reversal: an embedded arc of one contains an arc of the other, and uniqueness of the geodesic equation applies to their common tangent line. Reversal cannot preserve a single oriented phase orbit, since a fixed point of the parameter reflection would have zero velocity. Thus the critical levels are $m\ell_i$ with two orientations per label, locally finitely many. A fixed-width interval contains uniformly many labels, and there are $O(1+a)$ labels below a .

Let $I_{i,m}$ denote the Morse index. The periodic index form gives

$$|I_{i,m+k} - I_{i,m} - I_{i,k}| \leq n, \quad |I_{i,m} - m\Delta_i| \leq n, \quad \Delta_i \geq 0. \quad (11)$$

Indeed, identify the two form domains after imposing equality of the values at the joining point; each loses codimension at most n . The second inequality follows by subadditivity and superadditivity along multiples. In particular $|I_{i,m} - 2I_{i,m/2}| \leq n$ for even m .

The following uniform local facts will be used. There are constants A, B such that the ordinary group of one oriented critical circle has total rank at most B and is supported within distance A of $I_{i,m}$. Its reflection block is this ordinary group. Its K -block is induced from the half-shift action on one orientation. For even m its fixed block has root support within distance A of $I_{i,m/2}$; for odd m its fixed block is empty. Cup product with e is zero on every reflection crossing. The identifications commute with the actual restriction and connecting maps.

Here are the local constructions. On a sufficiently fine equal-time grid, replacing the central portion of each cell by its minimizing F -segment gives an energy-nonincreasing polygon retraction. It commutes with reversal and grid shifts. Continuity when the replaced interval shrinks follows by bounding its energy by the original energy on that interval. The discrete energy is

$$f_N(x_0, \dots, x_{N-1}) = N \sum_j d_F(x_j, x_{j+1})^2.$$

It is $C^{1,1}$, and smooth near each positive critical polygon. At a critical vertex the incoming and outgoing Legendre momenta agree. A zero edge would force the adjacent edge to be zero, by strict convexity, and hence force all edges to vanish. The index form splits into its piecewise Jacobi part and a positive part vanishing at the vertices, so the polygon model preserves indices and nullities.

Let P_i be the Jacobi monodromy, and choose D_i divisible by the orders of all its root-of-unity eigenvalues. For $s = \gcd(m, D_i)$,

$$\ker(P_i^m - 1) = \ker(P_i^s - 1).$$

On each contributing Jordan block the relevant root of $z^m - 1$ is simple. The characteristic-group iteration isomorphism for equal nullity therefore reduces the characteristic groups to finitely many roots $s \mid D_i$. The splitting of the energy into negative, null and positive directions retains the index shift $I_{i,m} - I_{i,s}$. The null slice has dimension at most $2n - 2$; the two-arc cover of the critical circle gives total rank at most four times the largest root characteristic rank and support in $[I_{i,m}, I_{i,m} + 2n - 1]$. These statements also include $\Delta_i = 0$, when (11) gives $I_{i,m} \leq n$.

Use one invariant descending flow for all pairs and their fixed loci. A localized field $-\nabla f_N / (1 + |\nabla f_N|^2)$ is locally Lipschitz. On compact regular strips its decrease is bounded below. For two tight regular cuts, a sufficiently long flow time T gives the lens

$$X = \{f_N \leq b\}, \quad Y = \{f_N \circ \varphi_T \leq a\} \cap X, \quad N_0 = \{f_N \circ \varphi_T \geq a\} \cap X, \quad Q = N_0 \cap Y.$$

The flow and its continuous hitting times identify (X, X_-) with (X, Y) and the local lens pair (N_0, Q) in relative cohomology. Intersecting these same sets with the half-shift fixed locus gives

the actual fixed restriction. This also proves naturality for the crossing exact sequences and their label decomposition. To use smooth compact boundaries, approximate f_N in C^1 by one invariant smooth function h on all required regular collars. For $f_s = (1-s)f_N + sh$, an invariant transverse field W gives

$$V_s = -\frac{h - f_N}{df_s(W)}W.$$

Its isotopy preserves f_s along moving boundary points. Applying this one isotopy and then restricting to fixed loci preserves all the comparison maps. Only a collar homeomorphism is required.

Finally choose a point of S^n outside the finite union of prime images. All sufficiently small critical neighborhoods evaluate into the punctured sphere. Its top cohomology vanishes, so e acts as zero on these local reflection pairs, and hence on the crossing.

6 A bound on finite-cut defects

For a regular pair $(2a, a)$, the fixed-locus identification gives

$$H_K^J(X_{2a}^\tau, M) = \bigoplus_{l \leq J} v^{J-l} B_a^l.$$

Actual restriction to this locus is an isomorphism in sufficiently high degrees. To see this, double a compact regular polygon model along its boundary. Outside a tubular neighborhood of its fixed submanifold there is a finite invariant cover with stabilizers $1, R, \langle \tau r \rangle$. The class $U = v(v+w)$ is zero on each stabilizer. Relative cup products over the cover make a power of U kill the relative group of the complement. The original restriction is a retract of this restriction on the double, so it becomes an isomorphism after inverting U .

The class w vanishes on every ambient local block, since the block is induced from one orientation with its half-shift action. Crossing exactness makes w nilpotent at a finite cut, and likewise on the fixed group. Thus inverting U is equivalent to inverting v . The sign sequence for $K \rightarrow R$ makes multiplication by v an isomorphism above the local index support, and eventually at each finite cut. Clearing denominators then makes the original restriction an isomorphism in that range. For a local label this range is $q \geq I_{i,m} + A$, after enlarging A using (11); both source and fixed target have already stabilized. No uniform bound on the denominator itself is required.

Put $\mathcal{V}_a = \bigoplus_l B_a^l$ and $\mathcal{H}_J = H_K^J(X_{2a}, M)$. Drop the powers v^{J-l} in actual restriction and write

$$\epsilon_J : \mathcal{H}_J \rightarrow \mathcal{V}_a, \quad E_J = \text{im } \epsilon_J, \quad t_J = \dim \ker \epsilon_J.$$

Let P_j project to root labels $l \leq j$, and define

$$\begin{aligned} D_{j,q} &= \dim P_j \mathcal{V}_a - \dim P_j E_q, \\ H_{j,J} &= \dim(E_J \cap \ker P_j), \\ \Phi_j(a) &= D_{j,2j} + D_{j,2j-1} + H_{j,2j} + t_{2j}. \end{aligned}$$

Each term is nonnegative and vanishes for large j at a fixed cut. The image E_J need not be graded by root label.

Lemma 4. *There is C such that $\sum_{j \geq 1} \Phi_j(a) \leq C(1+a)$.*

Proof. At a crossing, prime the post-crossing groups and write C^q for the actual local K -group and $\mathcal{W}$ for the total fixed reflection group. The natural exact rows are

$$\mathcal{H}_{q-1} \rightarrow C^q \rightarrow \mathcal{H}'_q \rightarrow \mathcal{H}_q \rightarrow C^{q+1}, \quad \mathcal{V} \xrightarrow{\partial} \mathcal{W} \xrightarrow{f} \mathcal{V}' \xrightarrow{i} \mathcal{V} \xrightarrow{\partial} \mathcal{W}.$$

The vertical maps are actual restriction followed by normalization; the lower row is obtained by inverting v . The maps f, i preserve root labels and ∂ raises them by one.

Fix $J = 2j$. Outside a fixed index window $|I_m - 2j| \leq C_0$, the local restrictions in degrees $J - 1, J, J + 1$ are injective with image $P_j \mathcal{W}$, and $\mathcal{W}^j = \mathcal{W}^{j+1} = 0$. Set $P = P_j$, $L = P\mathcal{V}$, $L' = P\mathcal{V}'$, $A_0 = f(P\mathcal{W})$ and

$$r = \text{rank}(\partial|_L), \quad r_q = \text{rank}(\partial|_{PE_q}).$$

Exactness gives

$$\dim P\mathcal{W} = \dim A_0 + r, \quad \dim L' - \dim L = \dim A_0 - r.$$

For $z \in E_q$, naturality puts ∂z in $P\mathcal{W}$. The low and high parts cannot cancel across the projection edge, so $\partial z = \partial(Pz)$. Target injectivity and actual exactness then give $i(PE'_q) = PE_q \cap \ker \partial$, with kernel A_0 . Consequently

$$D'_{j,q} - D_{j,q} = -(r - r_q) \quad (q = J - 1, J),$$

and the actual local exact row gives

$$(H_{j,J} + t_J)' - (H_{j,J} + t_J) = r - r_{J-1}.$$

Thus $\Phi'_j - \Phi_j = -(r - r_J) \leq 0$.

Inside the index window the local rank bounds instead bound every kernel and cokernel in these rows by a common M_0 . Using

$$\Phi_j = 2 \dim L - 2 \dim PE_J - \dim PE_{J-1} + \dim \mathcal{H}_J$$

gives $|\Phi'_j - \Phi_j| \leq 6M_0$. Only uniformly many integers j are inside the windows of the uniformly many labels at one event. The potentials start at zero, and there are $O(1 + a)$ events below $2a$. Summation proves the claim. $\square$

Define

$$Q_a^j = \text{coker}(B_\infty^j \rightarrow B_a^j), \quad T(a) = \sum_j \dim Q_a^j.$$

Lemma 5. *The nonnegative integer $T(a)$ is bounded independently of the positive regular cut a .*

Proof. Put $\beta_i = \Delta_i/\ell_i$, and list the distinct slopes $\{0, \alpha, \beta_i\}$ in increasing order. Outside fixed-width bands about $\beta_i a$, no ordinary local group occurs between a and a regular $a_+ \in (a + D, a + D + 1)$ in the relevant neighboring degrees. Lemma 1 therefore makes ordinary global restriction surjective there. By (3) it is an isomorphism below αa and its target is zero above it, after fixed margins are removed.

For consecutive slopes $s < s'$, choose one margin c large enough for all these comparisons. In the gap

$$sa + c < j < s'a - c$$

the map $Q_a^j \xrightarrow{w} Q_a^{j+1}$ is an isomorphism whenever both degrees lie in the gap. Below αa , apply the sign sequence to (Λ, X_a) : ordinary relative groups vanish in the gap, so w is invertible there. After a further n -degree margin, the image in B_∞ is divisible by w^n and is zero, giving the required quotient identification. Above αa , the ordinary groups of X_a vanish, so w is invertible on B_a . Its global image is killed by w^n , and hence is zero. Also $B_a^j = 0$ above the largest slope times a plus a constant.

Suppose that a gap contains a nonzero quotient class. Write $A = sa$, $r = (s' - s)a$, so $r \geq \delta a$ for a fixed $\delta > 0$. Choose a class y_{l_0} with nonzero quotient in degree $l_0 = \lceil A + 0.10r \rceil$ and put $y_l = w^{l-l_0} y_{l_0}$ up to degree $A + 0.90r$. Let

$$W_0 = \text{span}\{y_l : A + 0.15r \leq l \leq A + 0.20r\}, \quad q = \lfloor 0.50r \rfloor.$$

These classes and their w^q -multiples are independent by degree. By Lemma 4, some integer $k \in [A + 0.30r, A + 0.35r]$ has $D_{k,2k} \leq C_1$. Hence $W_0 \cap P_k E_{2k}$ has dimension at least $c_1 r$. Lift a basis to actual ambient classes. Their normalized images are $z_\mu + c_\mu$, with $z_\mu \in W_0$ and all labels in c_μ greater than k . Multiplication by w^q keeps the selected components independent, in labels at most $A + 0.70r$; the other components have labels greater than $A + 0.80r - 1$ or vanish. All remaining labels are greater than $A + 0.64r$ for large r .

For each $k' \in [A + 0.61r, A + 0.64r]$, multiply further by $v^{2k'-(2k+q)}$. This exponent is nonnegative because $2k + q \leq 2A + 1.20r < 2k'$. Multiplication by v leaves normalized root coefficients unchanged. Thus $H_{k',2k'} \geq c_1 r$ for at least $c_2 r$ integers k' . This contradicts the linear bound in Lemma 4, since $c_1 c_2 r^2 \geq c_1 c_2 \delta^2 a^2$.

Only uniformly many integer degrees near the finitely many slope bands can therefore contribute to $T(a)$ for large paired regular cuts. In high degrees the crossing rank bound gives

$$\dim B_a^j \leq C \sum_i \#\{m : m\ell_i < a, |j - m\Delta_i| \leq A + n\}.$$

Each summand is uniformly bounded if $\Delta_i > 0$, and is zero in high degrees if $\Delta_i = 0$. In the fixed low degrees, the ordinary global kernel is bounded by the uniformly many crossings in $(a, a + D + 1)$, by Lemma 1. The ordinary global rank is bounded too. Induction in (8), starting in negative degrees, bounds each low-degree B_a^j . This proves the required uniform bound. A cut with $2a$ critical can be perturbed within its regular interval so that both cuts are regular, without changing B_a or its global restriction. Bounded cuts contain finitely many events and cause no further difficulty. $\square$

7 Doubling

Write

$$V(a) = \sum_j \text{rank}(B_\infty^j \rightarrow B_a^j), \quad T(a) = \dim B_a^* - V(a).$$

For the basis of Lemma 2, let $b_{m,k}(a)$ indicate that $w^k X_m$ is nonzero at the cut. Let $h_{m,k}(a)$ indicate that $ew^k X_m$ is independent modulo all bottom images in its degree. Within either family the degrees are distinct. Thus

$$V(a) = \sum_{m,k} (b_{m,k}(a) + h_{m,k}(a)), \quad 0 \leq h_{m,k}(a) \leq b_{m,k}(a).$$

Use subscripts o, e on V for odd and even m .

Lemma 6. *If $2a$ is regular, $T(2a) \geq T(a)$. Equality implies $h_{m,k}(2a) = b_{m,k}(2a)$ for every odd m and every k .*

Proof. Let $H = H_G^*(X_{2a}, M)$. By (6), H/UH is finite-dimensional. Lift a homogeneous basis and induct down the degree to see that H is a finitely generated graded $\mathbb{F}_2[U]$ -module. Its torsion module $\mathcal{T}$ is finite-dimensional and killed by a power of U : a polynomial with nonzero constant term is injective by its lowest homogeneous component. The torsion-free quotient $\mathcal{F}$ has a homogeneous free basis, again by lifting a basis modulo U and using degree induction. Put

$$t = \dim(\mathcal{T}/U\mathcal{T}) = \dim \ker(U|_H), \quad \rho = \text{rank}_{\mathbb{F}_2[U]} \mathcal{F}.$$

Summing (6) gives $\dim B_{2a}^* = \rho + 2t$. In two consecutive sufficiently high degrees the dimensions of H sum to ρ . Sequence (7) and high-degree actual fixed restriction identify this sum with $\dim B_a^*$. Hence

$$\dim B_{2a}^* = \dim B_a^* + 2t. \tag{12}$$

Put $N = \sum_{m \text{ odd}, k} b_{m,k}(2a)$. The corresponding global G -lifts are U -torsion by Lemma 2, and their images in H/UH are independent. Thus $t = N + J_0$ with $J_0 \geq 0$. The selected even bottom and top lifts are independent modulo U together with those odd bottoms. Their projections to $\mathcal{F}/U\mathcal{F}$ therefore have rank at least $V_e(2a) - J_0$. Multiplying a homogeneous independent family by powers of U puts it into two consecutive large degrees without losing independence. The ambient global-image ranks in those degrees have sum at least $V_e(2a) - J_0$.

The half-shift fixed locus is G -invariant. Actual G -restriction to it is injective in high degrees, by (7) and the actual K -restriction isomorphism. On this fixed locus rotations act with doubled parameter. Denote the fixed global-image spaces by I_G^q, I_K^q . Naturality in (7) gives a surjection $I_K^D \rightarrow I_G^{D-1}$, since the global push is surjective; the injected I_G^D lies in its kernel. Therefore

$$\dim I_G^D + \dim I_G^{D-1} \leq \dim I_K^D = V(a)$$

for large D . The last equality is coefficientwise restriction $B_\infty[v] \rightarrow B_a[v]$. We obtain $V(a) \geq V_e(2a) - J_0$. Subtracting global-image ranks from (12) gives

$$\begin{aligned} T(2a) - T(a) &= 2(N + J_0) - V_o(2a) - V_e(2a) + V(a) \\ &\geq J_0 + \sum_{m \text{ odd}, k} (b_{m,k}(2a) - h_{m,k}(2a)) \geq 0. \end{aligned}$$

Equality forces each displayed summand to vanish. □

8 Conclusion

By Lemma 5, T is a bounded nonnegative integer on regular cuts, and it is constant between critical levels. Choose a bounded regular interval $I = (u, v)$ on which its maximum is attained. Every regular $x \in 2^k I$ has regular successive halves, because a critical half-level doubles to a critical level. Lemma 6 therefore forces $T(x)$ to be this maximum. For $k \geq 1$ equality holds between x and $x/2$ at every regular point of $2^k I$.

The odd bottom degrees form the progression

$$j_t = (4t + 1)d, \quad t \geq 0.$$

Choose $k \geq 1$ large enough that

$$\alpha 2^k(v - u) > 2\mathcal{C} + 4d, \quad \alpha 2^k u + \mathcal{C} > 2n + 1.$$

Some j_t then lies strictly between $\alpha 2^k u + \mathcal{C}$ and $\alpha 2^k v - \mathcal{C}$. Let $\xi = X_{2t+1}$. By Lemma 3, there are regular $x_1 < x_2$ in $2^k I$ such that ξ is zero at x_1 and nonzero at x_2 . Visibility is monotone with increasing cut, since restriction to a smaller sublevel factors through a larger one. Only finitely many crossings occur between x_1 and x_2 .

At the first crossing where ξ becomes nonzero, choose tight regular cuts $x_- < x_+$ still in $2^k I$. Exactness puts $\xi|_{X_{x_+}}$ in the image of

$$H_R^{j_t}(X_{x_+}, X_{x_-}) \longrightarrow B_{x_+}^{j_t}.$$

Cup product with e is zero on this crossing. Its naturality gives

$$\xi|_{X_{x_+}} \neq 0, \quad (e\xi)|_{X_{x_+}} = 0.$$

Thus $b_{2t+1,0}(x_+) = 1$ and $h_{2t+1,0}(x_+) = 0$. But $T(x_+) = T(x_+/2)$, so Lemma 6 requires these two indicators to agree. This contradiction excludes a finite set of prime geodesic images and proves Theorem 1. □